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Jeremy A. Collins

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EDUCATION

Georgia Institute of Technology **Atlanta, GA**

- Candidate for Master of Science in Robotics *08/2021 – 05/2023*
- GPA: 4.00/4.00
- Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, ML with Limited Supervision, Reinforcement Learning

University of Arkansas **Fayetteville, AR**

- B.S. in Mechanical Engineering, Minor in Mathematics *08/2017 – 05/2021*
- GPA: 4.00/4.00

PUBLICATIONS

- *Visual Estimation of Fingertip Pressure on Diverse Surfaces using Easily Captured Data* – Patrick Grady, **Jeremy A Collins**, Christopher D Twigg, Chengcheng Tang, James Hays, Charles C Kemp, *arXiv*. Link
- *Force/Torque Sensing for Soft Grippers using an External Camera* – **Jeremy A Collins**, Patrick Grady, Charles C Kemp, *IEEE International Conference on Robotics and Automation (ICRA) 2023*. Link
- *Visual Pressure Estimation and Control for Soft Robotic Grippers* – Patrick Grady, **Jeremy A Collins**, Samarth Brahmbhatt, Christopher D Twigg, Chengcheng Tang, James Hays, Charles C Kemp, *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2022*. Link
- *Tendon-Driven Soft Robotic Gripper for Blackberry Harvesting* – Anthony L Gunderman, **Jeremy A Collins**, Andrea L Myers, Renee T Threlfall, Yue Chen, *IEEE Robotics and Automation Letters (RA-L) 2022*. Link
- *Design and Control of a Peristaltic Pump to Simulate Left Atrial Pressure in a Conductive Silicone Model* – **Jeremy A Collins**, *Controls and Control Theory Commons 2021, Signal Processing Commons 2021*. Link

PATENTS

- *Soft Robotic Gripper for Berry Harvesting* - Yue Chen, Anthony L Gunderman, **Jeremy A Collins**. Link

SKILLS

- **Languages:** Python, C, C++, Arduino, MATLAB, Java
- **Tools:** PyTorch, Keras, NumPy, OpenCV, Pandas, ROS, OpenAI Gym, Git, COMSOL
- **Other Skills:** Competitive Rubik's Cube solver, trumpet player

TECHNICAL EXPERIENCE

Healthcare Robotics Lab **Atlanta, GA**
Graduate Research Assistant under Dr. Charlie Kemp *10/2021 – Present*

- Devised method to visually estimate robot contact pressure from a single RGB image using a deep neural network (collaboration with Meta Reality Labs)
- Led project to create vision-based force/torque sensor using ML/DL/CV techniques. Demos included vision-based robot fabric manipulation and cleaning complex surfaces
- Applied semi-supervised learning and domain adaptation methods to estimate fingertip contact pressure on curved and compliant surfaces using weakly labeled data (collaboration with Meta Reality Labs)

Dorabot Inc. **Atlanta, GA**
Engineering Intern *07/2021 – 05/2022*

- Collected, labeled, and curated image datasets for instance segmentation algorithms
- Designed and implemented robotic sorting systems for companies such as DHL
- Programmed simulations of robotic systems to demonstrate return on investment to clients
- Generated CAD models of system layouts to present to clients using SolidWorks

Medical and Soft Robotics Lab **Fayetteville, AR**
Undergraduate Research Assistant under Dr. Yue Chen *09/2019 – 06/2021*

- Devised and implemented drive system for blackberry harvesting soft gripper
- Created novel peristaltic pump with adaptive pressure control to simulate human heart

Marshalltown Company **Fayetteville, AR**
Mechanical Engineering Intern *05/2019 – 08/2019*

- Designed and developed products using Autodesk Inventor while considering costs and consumer preferences
- Decreased failure rate of robotic welding cell, increasing production speed of $3M in products annually
- Quantified manufacturing defect on high-priority product, increasing production by 75%

Joint Mathematics Meetings (JMM) **San Diego, CA**
Independent Researcher *01/2018*

- Exhibited for 5000+ attendees at the largest annual mathematics conference in the world

SELECTED PROJECTS

Video Prediction using Latent Diffusion *09/2022 – 12/2022*

- Used a transformer to predict frame sequence embeddings from Stable Diffusion encoder
- Predicted embedding sequences are denoised and decoded to produce future frames conditioned on text and video input

Learning Robotic Tasks from Video Observation *09/2022 – 12/2022*

- Designed robotic system to learn control policies using video data of expert demonstrations in OpenAI Gym with a causally masked transformer

Multi-Agent Reinforcement Learning *02/2022 – 05/2022*

- Implemented RL methods (Actor-Critic, PPO) to explore human-robot interaction in everyday activities
- Simulated game-theoretic interactions between humans and robots in simulation using OpenAI Gym

Distracted Driver Detection *02/2022 – 04/2022*

- Created system to classify driver behavior using PyTorch
- Employed multiple supervised (CNN, SVM) and unsupervised (PCA, clustering) machine learning methods

Midnight Stretch: A Semi-Autonomous Robotic Caregiver *08/2021 – 12/2021*

- Created a robotic system to autonomously detect falls in older adults from video using a deep neural network
- Features included LiDAR object segmentation, voice activation, and a web app for remote control and video calls

COMMUNITY INVOLVEMENT

- Member, AI Safety Initiative at Georgia Tech *09/2022 – Present*
- Attendee, PyTorch Conference 2022 *12/2022*
- Conference and workshop attendee, 2022 Conference on Neural Information Processing Systems (NeurIPS) *11/2022 – 12/2022*
- Attendee, Existential Risk Workshop, Global Challenges Project *11/2022*
- Judge, HackGT 2022 *10/2022*
- Mentor and judge, HackMIT 2022 *10/2022*
- Member, IEEE Power Electronics Society *09/2018 – 05/2021*
- Member, American Society of Mechanical Engineers (ASME) *05/2018 – 05/2021*
- President and founder, UArk Speedsolving Club (Rubik's Cube Club) *08/2017 – 05/2021*
    - Organized Rubik's Cube competition with 150+ in attendance *09/2018*
- Vice President, UArk Mathematics Club *08/2019 – 12/2019*

HONORS/AWARDS

- Coauthor of funded proposal ($60k) from AI Institute for Collaborative Assistance and Responsive Interaction for Networked Groups (AI-CARING) *10/2022*
- First-Ranked Senior Scholar, University of Arkansas *05/2021*
- Honors College Fellow, University of Arkansas *08/2017 – 05/2021*
- Arkansas Governor's Distinguished Scholar *08/2017 – 05/2021*
- Southern US Champion – Rubik's Cube Fewest Moves Challenge *2019-2021*
- Best Poster, 2018 FEP Honors Innovation Symposium (Track 6) *04/2018*
- MAA student travel grant for Joint Mathematics Meetings *01/2018*
- 1st place, Fall 2017 J.B. Hunt Hackathon *11/2017*